

Opportunistic Aortic Aneurysm Detection from Routine CT Scans

Marrowen is an offline AI prototype that automatically measures aortic diameter on routine non-contrast abdominal CT scans, flagging abdominal aortic aneurysm (AAA) without any additional imaging or patient burden. Built and validated entirely within the Kaggle notebook environment using freely available public datasets.

55 mm	0.97+	80%	Kaggle
Surgical threshold for repair	Published aorta segmentation Dice	Rupture mortality without detection	Training & validation platform

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Status Research Prototype — Not for Clinical Use

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Environment Kaggle Notebook (T4 GPU · 16 GB VRAM)

Regulatory Not FDA / CE / CDSCO cleared

Datasets All CC BY 4.0 or equivalent open license

01 · CLINICAL MOTIVATION

The most dangerous finding hiding in plain sight

Abdominal aortic aneurysm (AAA) is a silent enlargement of the aorta — the body's largest artery — caused by progressive weakening of the arterial wall. The condition produces no pain, no symptoms, and no warning until rupture. When rupture occurs, approximately 80% of patients die before reaching the operating table.

The tragedy is not that these patients were unseen — it is that they were scanned. Studies consistently show that patients who later die from AAA rupture had the aorta clearly visible in routine abdominal CT scans taken years earlier: for kidney stones, cancer follow-ups, unexplained abdominal pain. The aorta was there. The diameter was never measured, because measuring it was not the reason for the scan.

Marrowen closes this gap. Every routine abdominal CT that images the aorta contains a direct, unambiguous geometric measurement waiting to be taken. Aortic diameter in millimetres is not a learned proxy or a statistical correlation — it is a physical dimension. Marrowen measures it automatically, on scans already being performed, requiring no change to imaging protocols and no additional patient burden.

WHY CT IS IDEAL
• Aorta clearly delineated on non-contrast abdominal CT
• Diameter is a direct geometric measurement — no learned proxy
• TotalSegmentator segments aorta with Dice >0.97 out-of-the-box
• Non-contrast CT is the most common abdominal scan in emergency settings
• No gadolinium, no biopsy, no cost beyond the scan already performed
• Growth rate trackable across serial scans — longitudinal diameter delta
WHY COMPETITION IS LOW
• No FDA-cleared AI targets opportunistic AAA on routine CT
• All existing aortic AI operates on dedicated contrast CTA only
• Commercial PACS AI focuses on PE, fractures, nodules — not geometry
• Ultrasound screening covers only men 65+, single visit
• Academic models exist but none is packaged as a deployable tool
• Opportunistic angle on non-vascular CT entirely unaddressed commercially

02 · SCOPE BOUNDARY

What v1 delivers — and what it defers

IN SCOPE — v1 delivers this	OUT OF SCOPE — deferred to v2+
✓ Aortic segmentation from non-contrast abdominal CT (TotalSegmentator)	✗ Thoracic or thoracoabdominal aortic aneurysm (different FOV — v2)
✓ Maximum outer-wall aortic diameter measurement per axial slice	✗ Iliac artery aneurysm detection
✓ Anatomical location of widest point (vertebral level L1–L5)	✗ Contrast-enhanced CT correction / dual-energy virtual non-contrast
✓ Four-tier classification: Normal / Small AAA / Medium AAA / Large AAA	✗ DICOM network listener and PACS / RIS integration
✓ Automatic contrast-series filter via DICOM metadata + HU sanity gate	✗ FHIR R4 / HL7 output for EHR systems
✓ Growth rate delta if a prior scan is available for the same patient	✗ Real-time REST API inference service
✓ CSV results table + 1-page PDF report per patient	✗ Multi-site clinical deployment and regulatory documentation
✓ Validation on two separate public datasets (TotalSegmentator + AMOS 2022)	✗ Aortic dissection or wall thrombus detection
✓ Benchmark accuracy vs. radiologist manual diameter measurement (ICC >0.97)	

03 · DISEASE & IMAGING BIOMARKER

Why the measurement is the diagnosis

Unlike most AI diagnostic tasks, AAA detection does not require a learned classifier trained to recognise subtle visual patterns. The aortic diameter in millimetres — compared against a fixed clinical threshold — is the diagnosis. This makes the biomarker exceptionally clean: accuracy is limited primarily by the quality of the segmentation, not the complexity of the downstream model.

The aortic wall is clearly visible on non-contrast CT as a higher-attenuation ring. Outer-wall-to-outer-wall diameter, measured on axial slices and corrected for obliquity, is the clinical standard. Marrowen automates exactly this measurement.

Finding	Max Diameter	Annual Rupture Risk	Recommended Action
Normal	Below 30 mm	Negligible	Note in report — no action required
Small AAA	30 – 44 mm	Low (<1% / year)	Ultrasound surveillance every 2–3 years
Medium AAA	45 – 54 mm	Moderate (~1–3% / yr)	Vascular referral — surveillance q6–12 months
Large AAA	55 mm+	High (>5% / year)	Urgent vascular referral — surgical repair discussion

Contrast caveat

Intravenous contrast changes aortic wall conspicuity and invalidates outer-wall diameter measurements. Marrowen v1 filters out contrast-enhanced series by reading DICOM SeriesDescription and ProtocolName tags. A secondary HU sanity check (aortic lumen HU > 180 = flag as contrast) provides a redundant gate. All excluded series are logged in the output report with the reason code.

04 · PIPELINE — THREE STEPS

The simplest viable architecture

Marrowen v1 runs as a batch script. A folder of DICOM or NIfTI files is input; the three steps below execute per scan; results are written to the output directory next to the script. No server, no PACS, no network — clean, auditable Python.

01	Ingest & Filter pydicom reads DICOM metadata. Series flagged as contrast-enhanced (SeriesDescription / ProtocolName tags) or non-abdominal are excluded. A secondary HU gate (lumen HU > 180) provides redundant contrast detection. Remaining series are converted to NIfTI via dcm2niix and resampled to 1.5×1.5×5 mm. pydicom · dcm2niix · SimpleITK
02	Aorta Segmentation & Measurement TotalSegmentator v2 extracts the full aortic volume (published Dice >0.97). An ellipse is fitted to each axial cross-section of the mask using scikit-image regionprops. Outer-wall diameter = max(major axis) across all slices. The vertebral level of the widest point is recorded by co-registering the spine mask. If a prior scan exists, diameter delta is computed. TotalSegmentator v2 · NumPy · scikit-image
03	Classify & Report Maximum diameter is compared against the four clinical thresholds. Grade and recommended action are assigned deterministically. An XGBoost confidence layer flags measurement uncertainty (poor segmentation, non-standard FOV, heavy calcification). CSV row + 1-page PDF written per patient. XGBoost · ReportLab · pandas

Feature set going into XGBoost uncertainty layer

Feature	Role
Max outer-wall diameter (mm)	Primary output — directly compared to threshold
Vertebral level of max diameter	Location descriptor; flags thoracic misclassification
Aorta segmentation confidence	TotalSegmentator internal softmax score
Aortic mask volume (cm ³)	Sanity check — flags truncated or partial FOV
HU SD within aortic wall mask	Proxy for calcification artefact extent
Scanner kVp + manufacturer	Protocol correction for HU sanity gate
Patient age + sex (if in DICOM)	Risk context for report; not used in primary Dx

05 · TRAINING DATASETS

Freely available, no permissions required

Every dataset below is freely downloadable without institutional data-sharing agreements, ethics applications, or access approvals. All carry CC BY 4.0 or equivalent open licenses. Combined they provide over 1,700 CT volumes with expert aorta segmentation masks.

Dataset	N	Annotations	License	Kaggle Import
TotalSegmentator	1,228 CT volumes	117-class masks incl. full aorta + vertebrae; enables direct diameter extraction	CC BY 4.0	wasserth/totalsegmentator-dataset
AMOS 2022	500 CT volumes	15-organ masks incl. aorta; multi-site, multi-scanner; held out as external validation	CC BY 4.0	fudan-amos/amos22
LiTS 2017	131 CT volumes	Expert segmentation; clean non-contrast majority; aorta visible in majority of scans	CC BY 4.0	andreysemechko/lits17

PREPROCESSING	AUGMENTATION (TRAINING)
• Filter contrast series via DICOM metadata + HU gate (>180 HU = contrast)	• Random rotation $\pm 10^\circ$ all axes
• Resample to $1.5 \times 1.5 \times 5$ mm (TotalSegmentator default)	• Elastic deformation ($\sigma=8$, $\alpha=80$)
• HU clip: [-200, 500] — aortic window	• HU jitter ± 15 HU Gaussian noise
• Discard scans where aortic mask volume $< 20 \text{ cm}^3$ (truncated FOV)	• Random scaling 0.9–1.1×
• Vertebral co-registration for anatomical level annotation	• Random horizontal flip (left-right symmetry)

06 · TECHNOLOGY STACK

Minimal, entirely open-source

Layer	Library	Role	License
Segmentation	TotalSegmentator v2	Aorta + vertebral segmentation; no retraining required; Dice >0.97	Apache 2
Geometry	scikit-image 0.22	Ellipse fitting per axial slice; regionprops; outer-wall diameter	BSD
Classical ML	XGBoost 2.0 · scikit-learn 1.5	Measurement uncertainty flag; 5-fold CV; Platt calibration	Apache 2 · BSD
Image I/O	pydicom 2.4 · SimpleITK 2.3	Metadata filtering; resampling; HU extraction; NIfTI I/O	MIT · Apache 2
Conversion	dcm2niix	Batch DICOM → NIfTI; multi-frame and enhanced DICOM support	BSD
Data / Analysis	NumPy · pandas · SciPy	Diameter statistics; feature tables; results CSV	BSD
Reporting	ReportLab 4	1-page PDF per patient: grade, diameter, location, action	BSD
MLOps	MLflow 2.13	Experiment tracking; model registry; run comparison	Apache 2
Visualisation	Matplotlib	Aorta mask overlay PNGs; diameter-vs-slice plot for QA	BSD

Hardware & timing

TotalSegmentator inference (no retraining required): ~6 s per scan on T4 GPU, ~45 s on CPU. XGBoost uncertainty classifier training: < 5 min on CPU. End-to-end pipeline per scan: ~8 s on GPU, ~50 s on CPU. Full dataset preprocessing (1,700+ scans): ~4–6 hours on GPU.

07 · ACCURACY TARGETS & VALIDATION

Benchmark targets based on published literature

Because aortic diameter is a direct geometric measurement rather than a learned visual feature, accuracy is primarily limited by segmentation quality, not classifier complexity. TotalSegmentator's published Dice >0.97 sets a strong floor. The ICC target of >0.97 vs. radiologist manual measurement is supported by multiple published automated diameter studies using equivalent pipelines.

Task	Metric	Target	Basis
Aorta segmentation	Dice Similarity Coefficient	>0.97	TotalSegmentator published baseline
Diameter measurement	ICC vs. radiologist manual	>0.97	SliceOmatic + manual comparison studies
AAA detection (any)	AUROC	>0.95	Published automated CT diameter studies
4-class grading	Macro F1	>0.90	TotalSegmentator + AMOS combined benchmark
Contrast exclusion	Sensitivity	>0.99	DICOM metadata tag validation set
External generalisation	ICC drop (AMOS vs. LiTS)	<0.02 absolute	Multi-site published generalisability data
Diameter error	Mean absolute error	<2 mm	Clinical significance threshold for AAA surveillance

Why accuracy should be excellent

AAA detection reduces to a single geometric operation: fit an ellipse to the aortic cross-section and read the major axis. Even a rough segmentation with Dice 0.90 produces diameter measurements within 2–3 mm of manual measurement. TotalSegmentator at Dice >0.97 brings this error below 1 mm in most cases — well within clinical decision thresholds. The key treatment threshold interval is 10 mm wide (45–55 mm), making sub-2 mm accuracy more than sufficient for all four classification tiers.

08 · PROTOTYPE OUTPUT

What Marrowen v1 produces per scan

CT Series type	Non-contrast abdominal CT
Aorta segmentation Dice	0.971
Max outer-wall diameter	58.2 mm
Location	Infrarenal aorta · L2–L4
Surgical threshold	≥55 mm — surgical repair discussion
AAA grade	LARGE AAA
Prior scan delta	+4.1 mm / 14 months (rapid growth — flag)
Uncertainty flag	LOW — clean segmentation, no calcification artefact
Reason for scan	Kidney stone workup (opportunistic detection)

Artefacts generated per scan

PNG overlay	Aorta mask on representative axial slice with diameter annotation line
Diameter plot	Outer-wall diameter vs. axial slice number — QA visual for report
CSV row	Max diameter, location, grade, uncertainty flag, growth delta
1-page PDF	Grade badge, diameter plot, location string, recommended action
Exclusion log	Contrast-excluded scans logged separately with reason code

09 · BUILD PLAN — THREE SPRINTS

12 weeks, no infrastructure required

Sprint 1 Weeks 1–4 Data & Segmentation	• Import TotalSegmentator, AMOS, LiTS datasets • Run TotalSegmentator v2 on all scans; validate aorta masks visually (PNG overlays) • Contrast filter: test on 50 manually labelled series; confirm sensitivity >0.99 • Ellipse fitting pipeline: per-slice major axis extraction via scikit-image • Vertebral level co-registration for anatomical location annotation • MLflow tracking configured; all runs logged	Exit criterion Aorta Dice >0.97 on TotalSegmentator val split. Contrast filter sensitivity >0.99.
Sprint 2 Weeks 5–8 Measurement & Validation	• ICC vs. radiologist manual diameter on 50-case QA set • Diameter error analysis: mean absolute error per grade category • XGBoost uncertainty classifier: 5-fold CV on segmentation confidence features • Ablation: threshold-only vs. XGBoost-flagged vs. combined • Growth delta pipeline: serial scan diameter comparison for repeat-scan patients • SHAP values for uncertainty flag — per-case explainability for radiologist	Exit criterion ICC >0.97 vs. manual. MAE <2 mm across all grade categories.
Sprint 3 Weeks 9–12 Batch Pipeline & External Validation	• End-to-end notebook: folder of DICOMs in → CSV + PDF per patient out • External validation on AMOS 2022 (unseen site, multi-scanner) • Failure mode analysis: calcified aortas, obesity, non-standard FOV • PDF report template finalised with clinical advisor review • v1 technical report written; v2 scope drafted (thoracic extension, PACS)	Exit criterion 100 scans end-to-end without failures. ICC drop vs. AMOS <0.02 absolute.

10 · RISKS & MITIGATIONS

Known failure modes specific to aortic diameter measurement

Risk	Likelihood	Impact	Mitigation
Contrast CT miscategorised as non-contrast	Low	High — false normal	DICOM tag check + HU lumen gate (>180 HU = flag); dual-gate filter
Heavily calcified aortic wall	Medium	Medium — diameter overestimate	HU SD within mask flags calcification; report as “measurement uncertainty”
Truncated aortic FOV (chest CT, obese patient)	Medium	Medium — missed aneurysm	Discard if mask volume <20 cm³; flag as “incomplete FOV”
Tortuous aorta inflates diameter on oblique slices	Low	Medium — overestimate	Centreline straightening via skeletonisation (Sprint 2 option)
TotalSegmentator drift on non-standard scanners	Low	Low — AMOS validation catches this	AMOS held out as unseen multi-scanner test; retrain if Dice drops >0.02
Large dataset processing time on CPU only	Medium	Low — minor delay	Cache segmentation outputs; inference-only (no training); GPU strongly recommended

What success looks like

- Aorta Dice >0.97 on TotalSegmentator validation split
- Diameter ICC >0.97 vs. radiologist manual (50-case QA set)
- Mean absolute diameter error <2 mm across all size categories
- AAA detection AUROC >0.95 on internal validation
- ICC drop <0.02 on AMOS external multi-site test
- 100 scans processed end-to-end without pipeline failures; PDF per patient
- Contrast exclusion sensitivity >0.99 on manually labelled series